



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

**SECP3623
Database Programming
2025/2026 SEMESTER 1**

Lab Assignment 2

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Section	:	2

Part A: Data Preparation

Question A1

Create database and connections

```
> use clinicdb
< switched to db clinicdb
> db.createCollection("appointments")
< { ok: 1 }
> db.createCollection("loginSessions")
< { ok: 1 }
clinicdb>
```

Question A2

Create 10 documents into collections (appointments and loginSessions)

```
> db.appointments.insertMany([
  {
    appointmentId: "A001",
    patientName: "Jane Hafandi",
    doctor: "John Tse",
    department: "Cardiology",
    status: "COMPLETED",
    fee: 150,
    appointmentDate: ISODate("2025-01-10")
  },
  {
    appointmentId: "A002",
    patientName: "Theiviya Bhullar a/l Rajeswary",
    doctor: "Hafizuddin bin Nik Ashraf",
    department: "Dermatology",
    status: "PENDING",
    fee: 80,
    appointmentDate: ISODate("2025-01-12")
  },
  {
    appointmentId: "A003",
    patientName: "Yen Siaw Kaa",
    doctor: "Tse Jio Lao",
    department: "Orthopedics",
    status: "CANCELLED",
    fee: 0,
    appointmentDate: ISODate("2025-01-13")
  },
  {
    appointmentId: "A004",
    patientName: "Noor Hjh Norsyamimie Zulhelmi binti Che Zulfaqar",
    doctor: "John Tse",
    department: "Cardiology",
    status: "COMPLETED",
    fee: 200,
    appointmentDate: ISODate("2025-01-15")
  },
  {
    appointmentId: "A005",
    patientName: "Hafizuddin bin Nik Ashraf",
    doctor: "John Tse",
    department: "Cardiology",
    status: "COMPLETED",
    fee: 150,
    appointmentDate: ISODate("2025-01-16")
  },
  {
    appointmentId: "A006",
    patientName: "Theiviya Bhullar a/l Rajeswary",
    doctor: "Hafizuddin bin Nik Ashraf",
    department: "Dermatology",
    status: "PENDING",
    fee: 80,
    appointmentDate: ISODate("2025-01-17")
  },
  {
    appointmentId: "A007",
    patientName: "Yen Siaw Kaa",
    doctor: "Tse Jio Lao",
    department: "Orthopedics",
    status: "CANCELLED",
    fee: 0,
    appointmentDate: ISODate("2025-01-18")
  },
  {
    appointmentId: "A008",
    patientName: "Noor Hjh Norsyamimie Zulhelmi binti Che Zulfaqar",
    doctor: "John Tse",
    department: "Cardiology",
    status: "COMPLETED",
    fee: 200,
    appointmentDate: ISODate("2025-01-19")
  },
  {
    appointmentId: "A009",
    patientName: "Hafizuddin bin Nik Ashraf",
    doctor: "John Tse",
    department: "Cardiology",
    status: "COMPLETED",
    fee: 150,
    appointmentDate: ISODate("2025-01-20")
  },
  {
    appointmentId: "A010",
    patientName: "Theiviya Bhullar a/l Rajeswary",
    doctor: "Hafizuddin bin Nik Ashraf",
    department: "Dermatology",
    status: "PENDING",
    fee: 80,
    appointmentDate: ISODate("2025-01-21")
  }
])
```

```

{
  appointmentId: "A005",
  patientName: "Daniel Tan",
  doctor: "Bishan a/l Sumisha Sodhy",
  department: "Neurology",
  status: "PENDING",
  fee: 320,
  appointmentDate: ISODate("2025-01-18")
},
{
  appointmentId: "A006",
  patientName: "Mohamad Samsul bin Sam",
  doctor: "Tse Jio Lao",
  department: "Orthopedics",
  status: "COMPLETED",
  fee: 120,
  appointmentDate: ISODate("2025-01-20")
},
{
  appointmentId: "A007",
  patientName: "Nuur Hjh Muazzah Khusaini binti Amil Akif",
  doctor: "Hafizuddin bin Nik Ashraf",
  department: "Dermatology",
  status: "CANCELLED",
  fee: 0,
  appointmentDate: ISODate("2025-01-22")
},
{
  appointmentId: "A008",
  patientName: "Grace Chong",
  doctor: "Bishan a/l Sumisha Sodhy",
  department: "Neurology",
  status: "COMPLETED",
  fee: 350,
  appointmentDate: ISODate("2025-01-25")
},
{
  appointmentId: "A009",
  patientName: "Muhammed Syed Kamaruddin bin Wan Saifuddin",
  doctor: "John Tse",
  department: "Cardiology",
  status: "PENDING",
  fee: 180,
  appointmentDate: ISODate("2025-01-27")
},
{
  appointmentId: "A010",
  patientName: "Hj A'zizul bin Razemi",
  doctor: "Hafizuddin bin Nik Ashraf",
  department: "Dermatology",
  status: "COMPLETED",
  fee: 90,
  appointmentDate: ISODate("2025-01-30")
}
})
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6969cec9d34b80926ca85f3b'),
    '1': ObjectId('6969cec9d34b80926ca85f3c'),
    '2': ObjectId('6969cec9d34b80926ca85f3d'),
    '3': ObjectId('6969cec9d34b80926ca85f3e'),
    '4': ObjectId('6969cec9d34b80926ca85f3f'),
    '5': ObjectId('6969cec9d34b80926ca85f40'),
    '6': ObjectId('6969cec9d34b80926ca85f41'),
    '7': ObjectId('6969cec9d34b80926ca85f42'),
    '8': ObjectId('6969cec9d34b80926ca85f43'),
    '9': ObjectId('6969cec9d34b80926ca85f44')
  }
}

```

```
db.loginSessions.insertMany([
  {
    staffId: "S001",
    staffName: "T'ng Jie Sha",
    role: "Nurse",
    status: "ACTIVE",
    ipAddress: "192.168.1.10",
    loginTime: ISODate("2026-01-07T08:30:00Z")
  },
  {
    staffId: "S002",
    staffName: "John Tse",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "192.168.1.11",
    loginTime: ISODate("2026-01-07T08:45:00Z")
  },
  {
    staffId: "S003",
    staffName: "Nurul Hajjah Syafiqah Muhaimi binti Sukymi",
    role: "Admin",
    status: "INACTIVE",
    ipAddress: "192.168.1.12",
    loginTime: ISODate("2026-01-07T09:00:00Z")
  },
  {
    staffId: "S004",
    staffName: "Siaw Toh Tow",
    role: "Nurse",
    status: "ACTIVE",
    ipAddress: "192.168.1.13",
    loginTime: ISODate("2026-01-07T09:15:00Z")
  },
  {
    staffId: "S005",
    staffName: "Hafizuddin bin Nik Ashraf",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "192.168.1.14",
    loginTime: ISODate("2026-01-07T09:30:00Z")
  },
  {
    staffId: "S006",
    staffName: "V. T. Asha",
    role: "Receptionist",
    status: "INACTIVE",
    ipAddress: "192.168.1.15",
    loginTime: ISODate("2026-01-07T09:45:00Z")
  },
  {
    staffId: "S007",
    staffName: "Nor Rasyiqah Zulkafli",
    role: "Admin",
    status: "ACTIVE",
    ipAddress: "192.168.1.16",
    loginTime: ISODate("2026-01-07T10:00:00Z")
  },
  {
    staffId: "S008",
    staffName: "Tse Jio Lao",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "192.168.1.17",
    loginTime: ISODate("2026-01-07T10:15:00Z")
  },
],
```

```

{
  staffId: "S009",
  staffName: "Shamini Ponnusamy a/l Balachandra Sakadivan",
  role: "Nurse",
  status: "INACTIVE",
  ipAddress: "192.168.1.18",
  loginTime: ISODate("2026-01-07T10:30:00Z")
},
{
  staffId: "S010",
  staffName: "Bishan a/l Sumisha Sodhy",
  role: "Doctor",
  status: "ACTIVE",
  ipAddress: "192.168.1.19",
  loginTime: ISODate("2026-01-07T10:45:00Z")
}
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6969d233d34b80926ca85f45'),
    '1': ObjectId('6969d233d34b80926ca85f46'),
    '2': ObjectId('6969d233d34b80926ca85f47'),
    '3': ObjectId('6969d233d34b80926ca85f48'),
    '4': ObjectId('6969d233d34b80926ca85f49'),
    '5': ObjectId('6969d233d34b80926ca85f4a'),
    '6': ObjectId('6969d233d34b80926ca85f4b'),
    '7': ObjectId('6969d233d34b80926ca85f4c'),
    '8': ObjectId('6969d233d34b80926ca85f4d'),
    '9': ObjectId('6969d233d34b80926ca85f4e')
  }
}

```

Part B: Appointment Reporting

Question B1

Retrieve “COMPLETED” appointment with fee between 100 and 300. Display appointmentId, patientName, department and fee. Sort the department in ascending order and fee in descending order.

```

db.appointments.find({status:"COMPLETED", fee:{ $gte:100,$lte:300 }},
  {_id:0, appointmentId:1,patientName:1,department:1,fee:1})
  .sort({department:1,fee:-1})
{
  appointmentId: 'A004',
  patientName: 'Noor Hjh Norsyamimie Zulhelmi binti Che Zulfaqar',
  department: 'Cardiology',
  fee: 200
}
{
  appointmentId: 'A001',
  patientName: 'Jane Hafandi',
  department: 'Cardiology',
  fee: 150
}
{
  appointmentId: 'A006',
  patientName: 'Mohamad Samsul bin Sam',
  department: 'Orthopedics',
  fee: 120
}

```

Question B2

Show total number of appointments and total fees collected for each department and sort the output of total fees in descending order.

```
db.appointments.aggregate([
  {
    $group: {
      _id: "$department",
      totalAppointments: { $sum: 1 },
      totalFees: { $sum: "$fee" }
    }
  },
  {
    $sort: { totalFees: -1 }
  }
])
[
  {
    _id: 'Neurology',
    totalAppointments: 2,
    totalFees: 670
  },
  {
    _id: 'Cardiology',
    totalAppointments: 3,
    totalFees: 530
  },
  {
    _id: 'Dermatology',
    totalAppointments: 3,
    totalFees: 170
  },
  {
    _id: 'Orthopedics',
    totalAppointments: 2,
    totalFees: 120
  }
]
```

Part C: Clinic Security Monitoring

Question C1

Retrieve ACTIVE staff login and display staffName, role, ipAddress and loginTime. Then, sort the results by most recent login time first.

```
db.loginSessions.find({status:"ACTIVE"},{_id:0,staffName:1,role:1,ipAddress:1,loginTime:1}).
sort({loginTime:-1})
[
  {
    staffName: 'Bishan a/l Sumisha Sodhy',
    role: 'Doctor',
    ipAddress: '192.168.1.19',
    loginTime: 2026-01-07T10:45:00.000Z
  },
  {
    staffName: 'Tse Jio Lao',
    role: 'Doctor',
    ipAddress: '192.168.1.17',
    loginTime: 2026-01-07T10:15:00.000Z
  },
  {
    staffName: 'Nor Rasyiqah Zulkafli',
    role: 'Admin',
    ipAddress: '192.168.1.16',
    loginTime: 2026-01-07T10:00:00.000Z
  }
]
```

```
{
  staffName: 'Hafizuddin bin Nik Ashraf',
  role: 'Doctor',
  ipAddress: '192.168.1.14',
  loginTime: 2026-01-07T09:30:00.000Z
}
{
  staffName: 'Siaw Toh Tow',
  role: 'Nurse',
  ipAddress: '192.168.1.13',
  loginTime: 2026-01-07T09:15:00.000Z
}
{
  staffName: 'John Tse',
  role: 'Doctor',
  ipAddress: '192.168.1.11',
  loginTime: 2026-01-07T08:45:00.000Z
}
{
  staffName: "T'ng Jie Sha",
  role: 'Nurse',
  ipAddress: '192.168.1.10',
  loginTime: 2026-01-07T08:30:00.000Z
}
```

Question C2a

List all distinct IP addresses used by staff with ACTIVE sessions

```
db.loginSessions.distinct("ipAddress",{status:"ACTIVE"})
[
  '192.168.1.10',
  '192.168.1.11',
  '192.168.1.13',
  '192.168.1.14',
  '192.168.1.16',
  '192.168.1.17',
  '192.168.1.19'
]
```

Question C2b

The number of unique ACTIVE staff members

```
db.loginSessions.aggregate([
  { $match: { status: "ACTIVE" } },
  { $group: { _id: "$staffId" } },
  { $count: "activeStaff" }
])
{
  activeStaff: 7
}
```

Part D: Indexing & Data Maintenance

Question D1

Create index for department and status

```
> db.orders.createIndex({department:1,status:1})
< department_1_status_1
```

Question D2a

Update CANCELLED appointments to ARCHIVED

```
db.appointments.updateMany(  
  {status:"CANCELLED"},  
  { $set: { status:"ARCHIVED"}}  
)  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 0,  
  modifiedCount: 0,  
  upsertedCount: 0  
}
```

Question D2b

Delete INACTIVE session

```
db.loginSessions.deleteMany(  
  { status: "INACTIVE"}  
)  
{  
  acknowledged: true,  
  deletedCount: 3  
}
```